

Mass Percent of Carbon in Sodium Bicarbonate



Introduction

The *law of definite proportions* states that a compound always contains two or more elements combined in a fixed (specific) ratio by mass. In this experiment, the mass percent of carbon in sodium bicarbonate (baking soda) will be determined by measuring the mass of carbon dioxide gas produced in the following chemical reaction (equation).



Concepts

- Law of definite proportions
- Law of conservation of matter

Background

The French chemist, Joseph Louis Proust (1754–1826), was one of the first to observe that elements combine with one another in a definite mass ratio. Proust's experimental work helped form the law of definite proportions, (also called the law of definite composition). According to this law, there is a precise quantity of each component required in the formation of a specific compound—the *percent composition* is constant. This, in turn, implies that in the decomposition of a specific compound, a precise quantity of each component will be produced.

Materials

Sodium bicarbonate, NaHCO_3 , 1.5 g

Vinegar, CH_3COOH , 180 mL

Balance, centigram (0.01-g precision)

Beaker, 250-mL

Spatula

Weighing dish, 3

Safety Precautions

Wear chemical splash goggles, chemical-resistant gloves, and a chemical-resistant apron. Wash hands thoroughly with soap and water before leaving the laboratory. Please review current Material Safety Data Sheets for additional safety, handling, and disposal information.

Procedure

1. Pour 60 mL of vinegar (5% acetic acid, CH_3COOH) into a 250-mL beaker.
2. Place a weighing dish and the beaker containing the vinegar side-by-side on the balance.
3. Record the combined mass of the beaker, vinegar, and weighing dish in the Data Table.
4. Weigh out about 1.5 g (1.4–1.5 g) of sodium bicarbonate on the weighing dish. Record this value in the Data Table.
Note: The exact amount is not important as long as the precise mass is recorded in step 5.
5. Record the combined mass of the beaker, weighing dish, sodium bicarbonate, and vinegar in the Data Table.
6. Slowly and carefully add the sodium bicarbonate to the vinegar in the beaker in such a way as to prevent any foam produced by the reaction from overflowing the beaker.
7. When all the sodium bicarbonate has been added, gently swirl the beaker to make sure the reaction is complete, and to remove any remaining bubbles in the beaker.
8. Record the mass of the beaker, its contents, and the weighing dish in the Data Table.
9. Repeat steps 1–8 two more times and enter the data in the data table (Trial 2 and Trial 3).

Data Table

	Trial 1	Trial 2	Trial 3
Mass of beaker, vinegar, and weighing dish (g)			
Mass of beaker, weighing dish, vinegar, and sodium bicarbonate (g)			
Mass of sodium bicarbonate (g)			
Mass of beaker, its contents, and weighing dish after the reaction (g)			
Mass of carbon dioxide produced (g)			
Ratio of the mass of carbon to the mass of carbon dioxide			
Mass of carbon produced (g)			
Percent carbon in sodium bicarbonate sample			

Post-Lab Questions and Calculations *(Show all work on a separate sheet of paper.)*

1. Subtract the combined mass of the weighing dish, beaker, and its contents after the reaction from the combined mass of the beaker, weighing dish, sodium bicarbonate, and vinegar before the reaction to find the mass of carbon dioxide lost to the atmosphere. Record this in the Data Table.
2. Divide the atomic weight of carbon by the molecular weight of carbon dioxide to determine the ratio of the mass of carbon to the mass of carbon dioxide. Record this ratio in the Data Table.
3. Multiply the mass of carbon dioxide by the mass ratio (question 2) to determine the mass of carbon produced in each trial. Record the results in the Data Table.
4. Divide the mass of carbon produced by the mass of sodium bicarbonate to determine the percent carbon in sodium bicarbonate for each trial. Record the results in the Data Table. Calculate the average percent carbon in sodium bicarbonate.
5. Divide the atomic weight of carbon by the molecular weight of sodium bicarbonate and multiply by 100% to determine the theoretical mass percent of carbon in sodium bicarbonate.
6. Compare the average experimental mass percent of carbon in sodium bicarbonate to the theoretical value and calculate the percent error. The equation for percent error is

$$\frac{|\text{theoretical} - \text{experimental}| \text{ mass of carbon in sodium bicarbonate}}{\text{theoretical mass of carbon in sodium bicarbonate}} \times 100\%$$

Disposal

Please consult your instructor for appropriate disposal procedures.